Sources

Pantothenic acid is present in dried mushrooms, avocado, dried egg yolks and sunflower seeds in high amounts. Outer layer of whole grains contains the vitamin, but milling removes much of the pantothenic acid.

Function

- Pantothenic acid forms a part of coenzyme A which is an essential coenzyme for metabolising carbohydrates, fat and proteins.
- It is necessary to synthesise vitamin D, steroid hormones and red blood cells
- It also helps to boost the immunity

Absorption and storage

Free pantothenic acid is absorbed into intestinal cells via a saturable, sodium-dependent active transport system. However, in foods, most pantothenic acid is in the form of CoA or bound to acyl carrier protein (ACP). Since the intestinal cells can only absorb this vitamin in free pantathenic acid, it is converted into free panthothenic acid in intestine.

Deficiency

Pantothenic acid deficiency is very rare and seen only in cases of severe malnutrition. Pantothenic acid is found in many common foods and average diets are thought to have an adequate amount of it.

8.3 Vitamin C

Vitamin C is called as ascorbic acid. Ascorbic acid is an enediol-lactone of an acid with configuration similar to that of the sugar L-glucose.

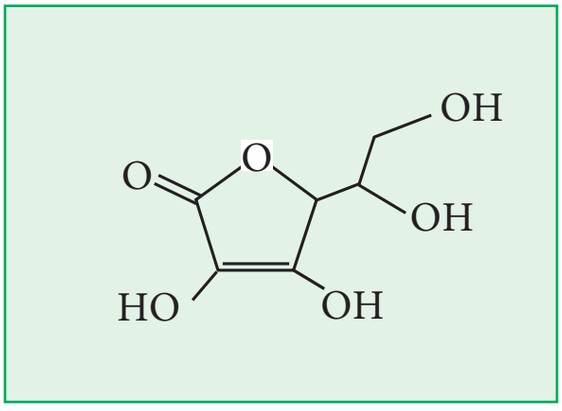

Fig. 8.29 Structure of vitamin C

Unit 8.indd 220 21-05-2018 15:11:47